Differences in Differences and neighbors

Stat 256: Causality

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The ID landscape

We covered:

- "Adjustment-based" strategies, i.e. those justified by backdoor criterion/CI.
- Instrumental variables, allowing unobserved confounding
- Regression Discontinuity, also allowing unobserved confounding

Difference-in-Difference:

- Allows a kind of unobserved confounding
- Requires "unconfoundedness of $Y(0)$ trend" instead, better known as "parallel trends"

DID in context

What if we think some unobservables that influence the potential outcomes may differ between treated and untreated units?

In some cases we can instead make assumptions about *how potential outcomes change over time*, and weaken the SOO assumption

- Differences in Differences (DID) approach is perhaps simplest way of using time-series data
- DID $\neq$ Two-way fixed effects (TWFE)
- Other ways to think about DID: placebo outcomes, relationship to sequential ignorability.

DID in the Time of Cholera

- Prevailing 19th century wisdom was that cholera was spread by “bad air”
- John Snow, a London physician thought otherwise
- In 1849, the Southwark and Vauxhall water company and the Lambeth company got water from the same polluted part of the Thames in central London. Snow assessed cholera mortality in both areas.
- In 1852, Lambeth moved its waterworks up stream.
- Shortly after, cholera death rates fell sharply in areas served by Lambeth. but not area served by S & V.

Intuitions:

- What assumptions allow us to draw a causal conclusion?
- Why not just compare the two in 1852?
- Why not just compare Lambeth’s service area post-minus-pre?

The DID Setup: Two Groups, Two Periods

Two groups:

- $D = 1$ for units that will take treatment at some point
- $D = 0$ for units that remain untreated

Two periods:

- $T = 0$ Pre-treatment period
- $T = 1$ Post-treatment period – but only some take it.

Potential outcomes:

- $Y_{di}(t)$ for treatment status d and time t.

DID Setup

Definition

(Contemporaneous) causal effect for unit i at time t is

$$\tau_{it} = Y_{1i}(t) - Y_{0i}(t)$$

Observed outcomes $Y_i(t)$ are realized as

$$Y_i(t) = Y_{1i}(t) \cdot (D_i * (t = 1)) + Y_{0i}(t) \cdot (1 - (D_i * (t = 1))$$

Estimand (ATT post-treatment)

$$\tau_{ATT} = \mathbb{E}[Y_{1i}(1) - Y_{0i}(1) | D_i = 1]$$

Two Groups and Two Periods

Estimand (ATT)

$$\tau_{ATT} = \mathbb{E}[Y_{1i}(1)|D_i = 1] - \mathbb{E}[Y_{0i}(1)|D_i = 1]$$

Problem

Missing potential outcome: $\mathbb{E}[Y_0(1)|D = 1]$

ID Strategy 1 (not DID):

- Use estimand: $\mathbb{E}[Y_1(1)|D = 1] - \mathbb{E}[Y_0(0)|D = 1]$
- What is the idea here? Under what conditions would it work?
- Post-minus-pre.
- Works when: $\mathbb{E}[Y_0(0)|D = 1] = \mathbb{E}[Y_0(1)|D = 1]$
- Why/when would this be wrong?

Two Groups and Two Periods

Estimand (ATT)

$$\tau_{ATT} = \mathbb{E}[Y_{1i}(1)|D_i = 1] - \mathbb{E}[Y_{0i}(1)|D_i = 1]$$

Problem

Missing potential outcome: $\mathbb{E}[Y_0(1)|D = 1]$

ID Strategy 2 (still not DID):

- Use: $\mathbb{E}[Y(1)|D = 1] - \mathbb{E}[Y(1)|D = 0]$
- What would you call this? When would it work?
- Cross-sectional comparison in post-period.
- Works if $\mathbb{E}[Y_0(1)|D = 0] = \mathbb{E}[Y_0(1)|D = 1]$
- Why/when would this be wrong?

Two Groups and Two Periods

Estimand (ATT)

$$\tau_{ATT} = \mathbb{E}[Y_{1i}(1)|D_i = 1] - \mathbb{E}[Y_{0i}(1)|D_i = 1]$$

Problem

Missing potential outcome: $\mathbb{E}[Y_0(1)|D = 1]$

ID Strategy 3: Difference-in-Differences (DID)

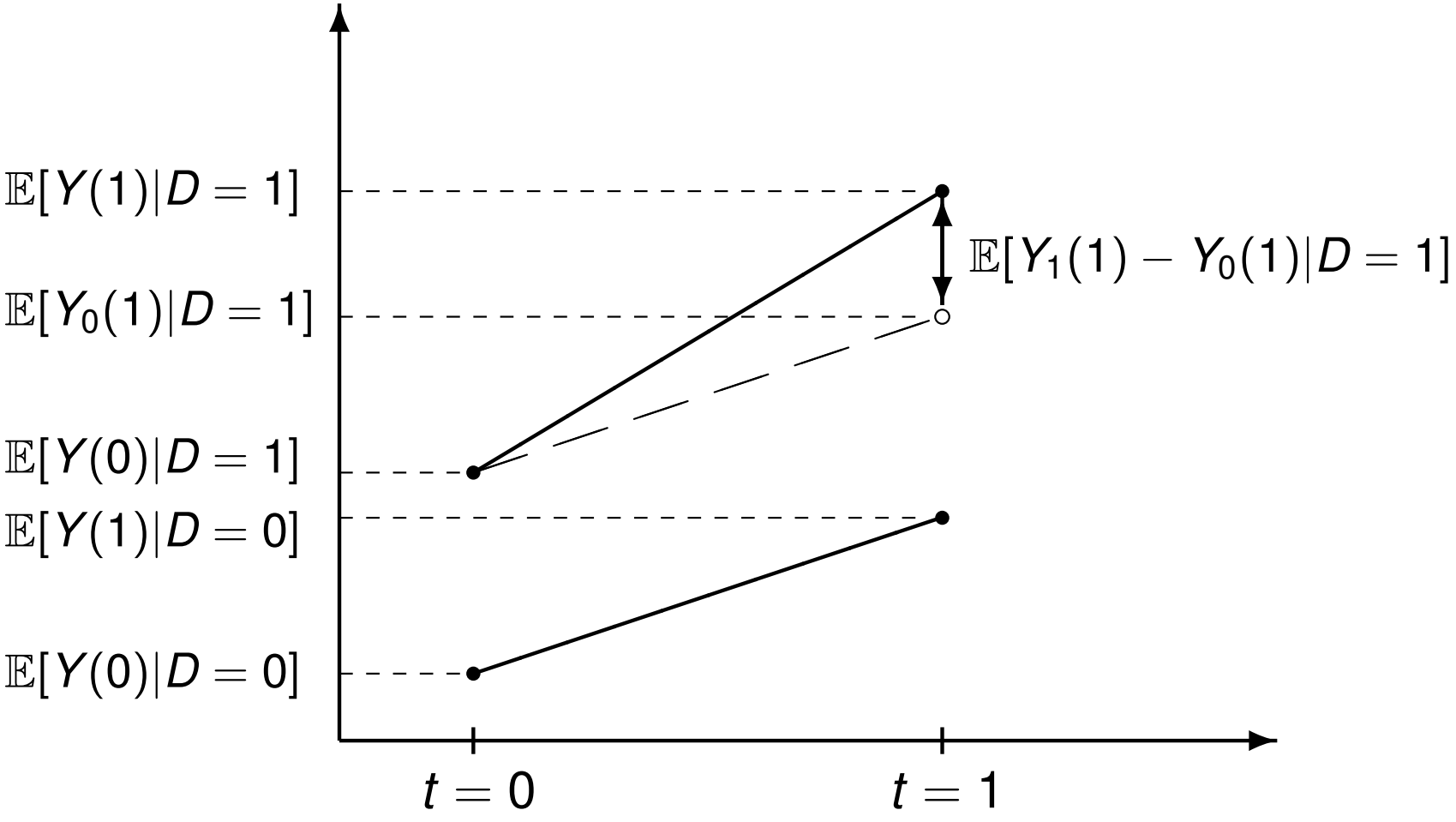
- Use:

$$\mathbb{E}[Y(1) - Y(0)|D = 1] - \mathbb{E}[Y(1) - Y(0)|D = 0]$$

- What is the idea? Assumes what?
- $\mathbb{E}[Y_0(1) - Y_0(0)|D = 1] = \mathbb{E}[Y_0(1) - Y_0(0)|D = 0]$

"Parallel trends" assumption

Graphical Representation: Difference-in-Differences



Identification with Difference-in-Differences

Identification Assumption (Parallel Trends)

$$\mathbb{E}[Y_0(1) - Y_0(0)|D = 1] = \mathbb{E}[Y_0(1) - Y_0(0)|D = 0]$$

Identification Result

Under parallel trends, the ATT is identified by the DID estimator:

$$\begin{aligned}\tau_{DID} &= \{\mathbb{E}[Y(1)|D = 1] - \mathbb{E}[Y(0)|D = 1]\} - \{\mathbb{E}[Y(1)|D = 0] - \mathbb{E}[Y(0)|D = 0]\} \\ &= \mathbb{E}[Y_1(1) - Y_0(1)|D = 1] = ATT\end{aligned}$$

Proof

$$
\begin{aligned}
\tau_{DID} &= \{\mathbb{E}[Y(1)|D=1] - \mathbb{E}[Y(0)|D=1]\} - \{\mathbb{E}[Y(1)|D=0] - \mathbb{E}[Y(0)|D=0]\} \\
&= \{\mathbb{E}[Y_1(1)|D=1] - \mathbb{E}[Y_0(0)|D=1]\} - \{\mathbb{E}[Y_0(1)|D=0] - \mathbb{E}[Y_0(0)|D=0]\} \\
&= \{\mathbb{E}[Y_1(1)|D=1] - \mathbb{E}[Y_0(0)|D=1]\} - \{\mathbb{E}[Y_0(1)|D=1] - \mathbb{E}[Y_0(0)|D=1]\} \\
&= \mathbb{E}[Y_1(1)|D=1] - \mathbb{E}[Y_0(1)|D=1] \\
&= ATT
\end{aligned}
$$

Remark on parallel trends:

Identification Assumption (Parallel Trends = First difference ignorability)

Conventionally we write,

$$\mathbb{E}[Y_0(1) - Y_0(0)|D = 1] = \mathbb{E}[Y_0(1) - Y_0(0)|D = 0]$$

But note that this is equivalent to:

$$\Delta Y_0 \perp D$$

where $\Delta Y_0 \equiv \boldsymbol{Y}_0(1) - \boldsymbol{Y}_0(0)$

Suggestion:

- Think of ΔY_{0i} as a characteristics of unit i and think about why/whether it would be independent of D.
- If you weren't able to argue that $Y_{0i} \perp D_i$, what supports the claim that $\Delta Y_{0i} \perp D_i$?

Difference-in-Differences: Estimators

Estimand (DID)

$$\tau_{DID} = \{\mathbb{E}[Y(1)|D=1] - \mathbb{E}[Y(1)|D=0]\} - \{\mathbb{E}[Y(0)|D=1] - \mathbb{E}[Y(0)|D=0]\}$$

Estimator

$$\frac{1}{N_1}\sum_{D_i=1}\{Y_i(1) - Y_i(0)\} - \frac{1}{N_0}\sum_{D_i=0}\{Y_i(1) - Y_i(0)\},$$

where N_1 and N_0 are the number of treated and control units respectively.

Threat 1: Non-parallel trends

Non-parallel trends: Often treatments/programs are targeted based on pre-existing *trend* in outcomes.

- Mean reversion/"Ashenfelter dip": treatment may be more likely to be taken up when there has been a recent drop in Y_0. Combined with a tendency to revert anyway, you get different Y_0 trends among treated and control.
- E.g. job training programs, trials for relapsing-remitting/ mean-reversing conditions.

Again, it's really claiming $\Delta Y_{0i} \perp D_i$, so challenge whether anything related to trend in outcome can relate to treatment probability.

Checks for Difference-in-Differences Design

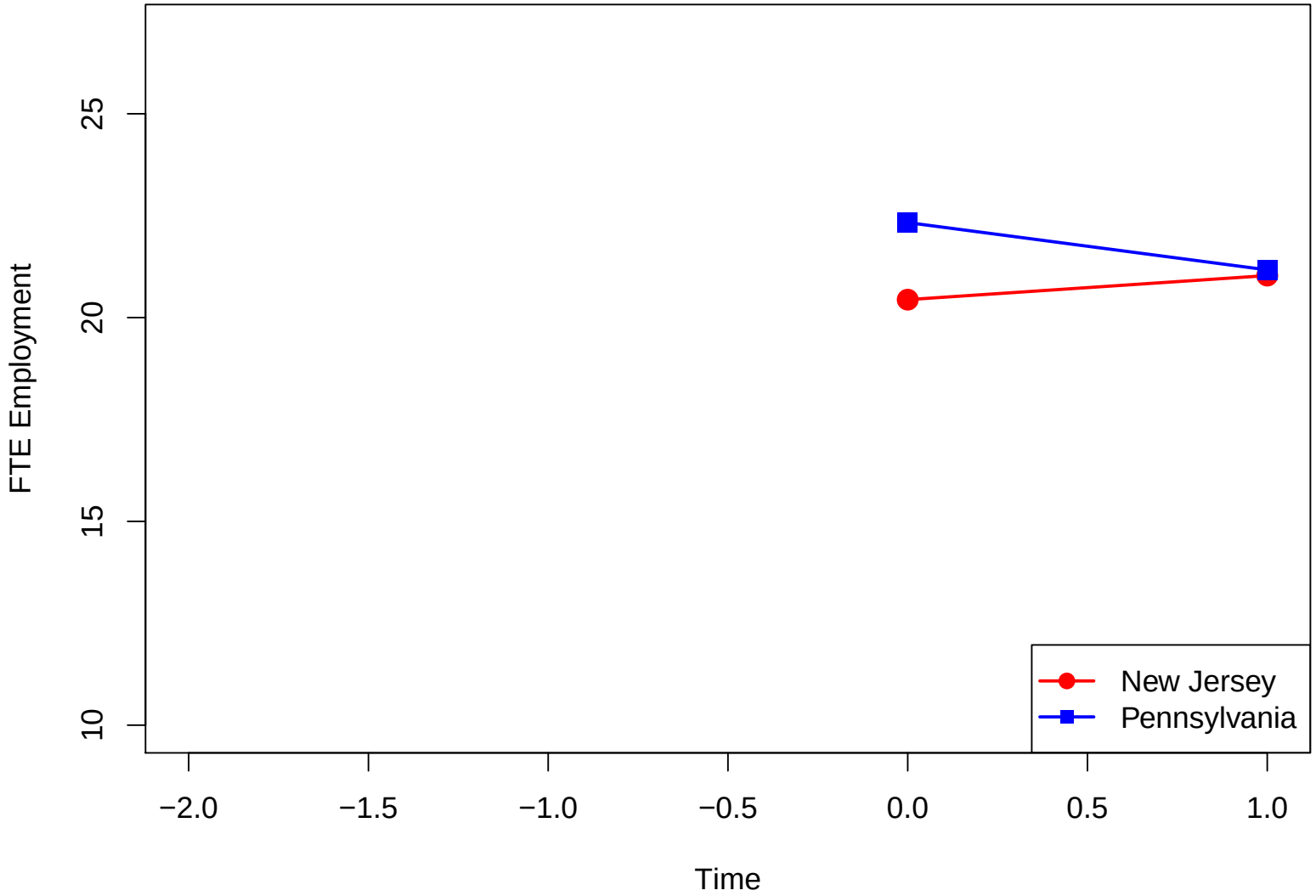
Two kinds of falsification tests for non-parallel trends

1. Falsification test using data for prior periods
   - Given parallel trends during periods $t = -1, 0, 1$, we have:

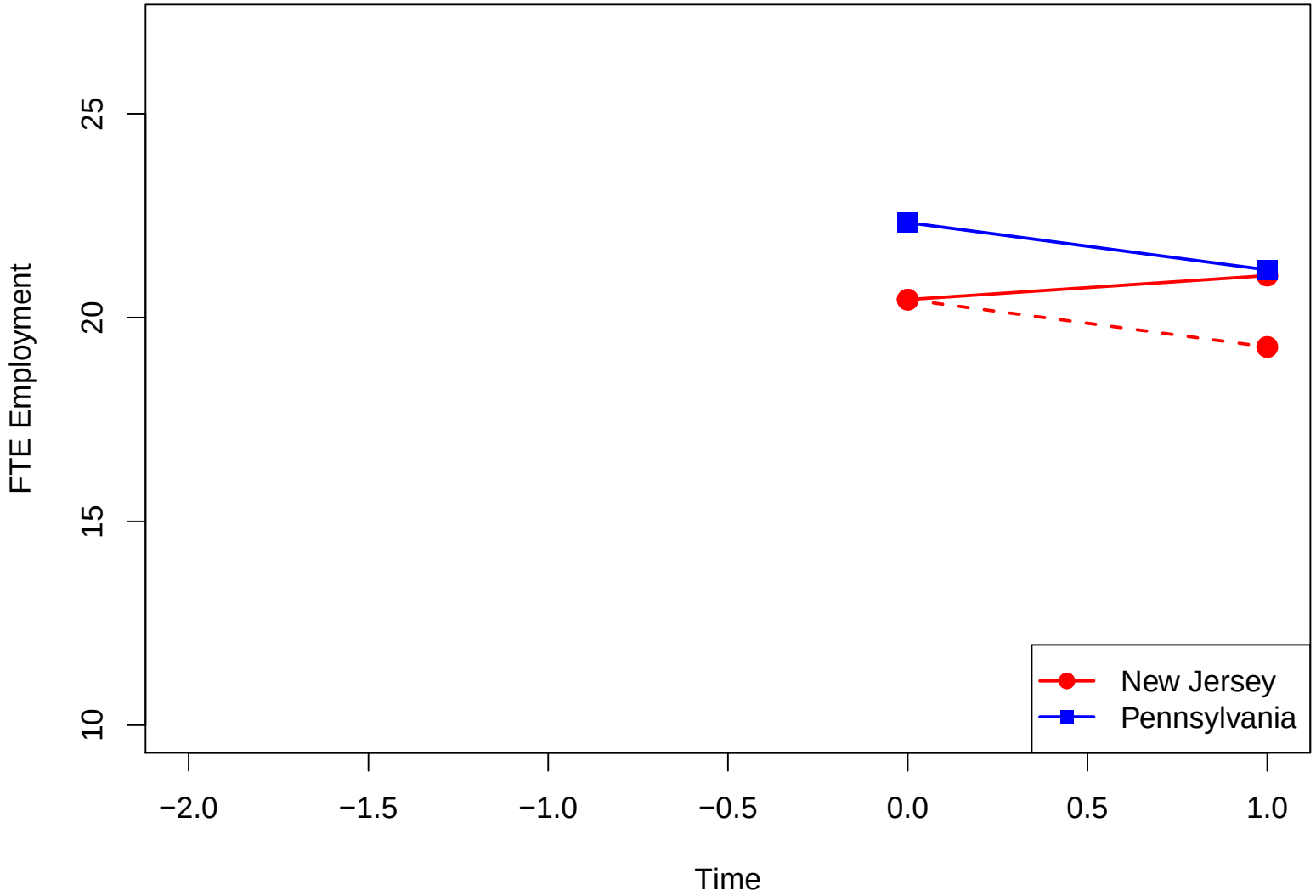
$$\mathbb{E}[Y(0) - Y(-1)|D = 1] - \mathbb{E}[Y(0) - Y(-1)|D = 0] = 0$$

   - run placebo DID on data from $t = -1, 0$.
2. Falsification test using alternative outcome or treated sub-group that is not supposed to be affected by the treatment
   - troubling if DID from the placebo outcome or un-movable group is non-zero.

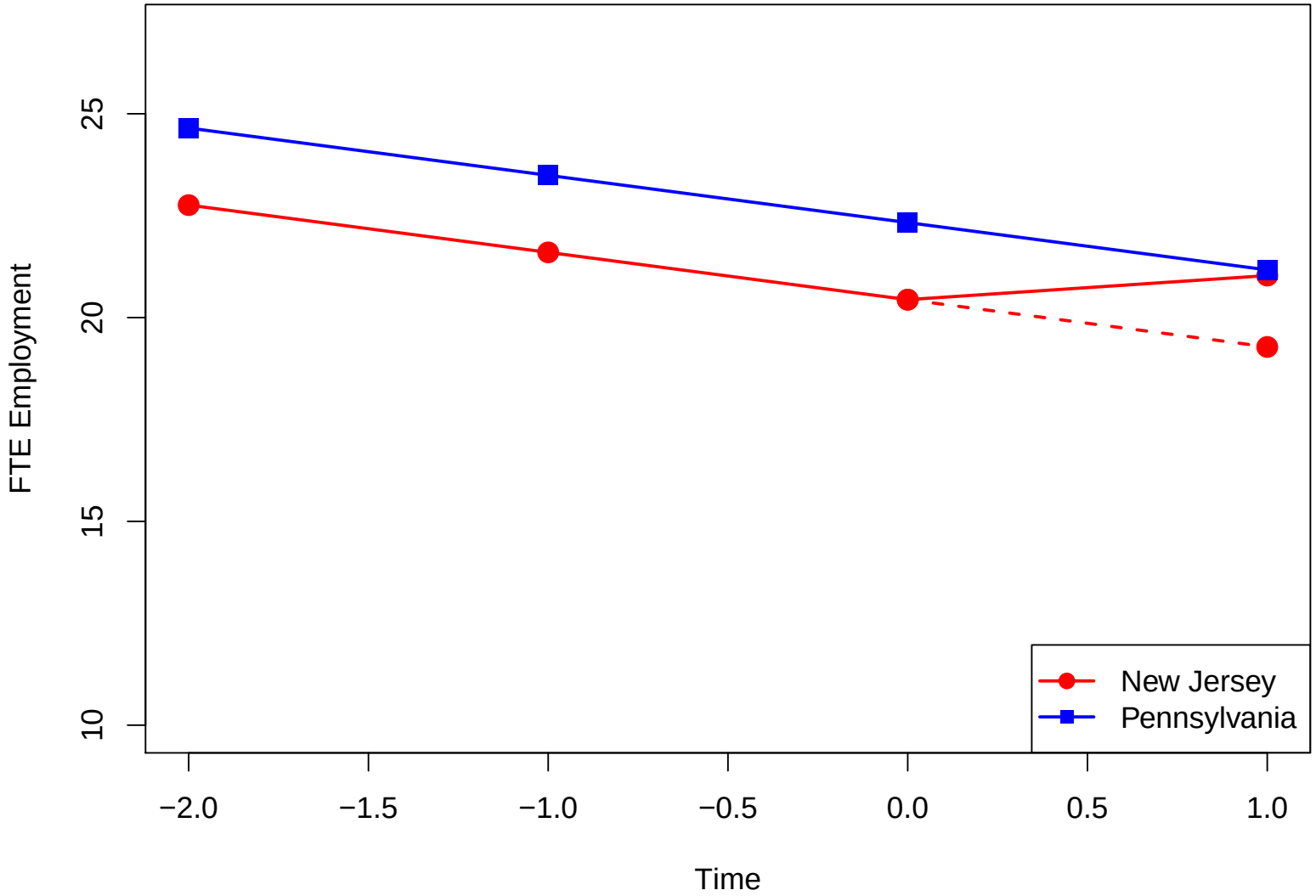
Falsification test: Data for prior periods



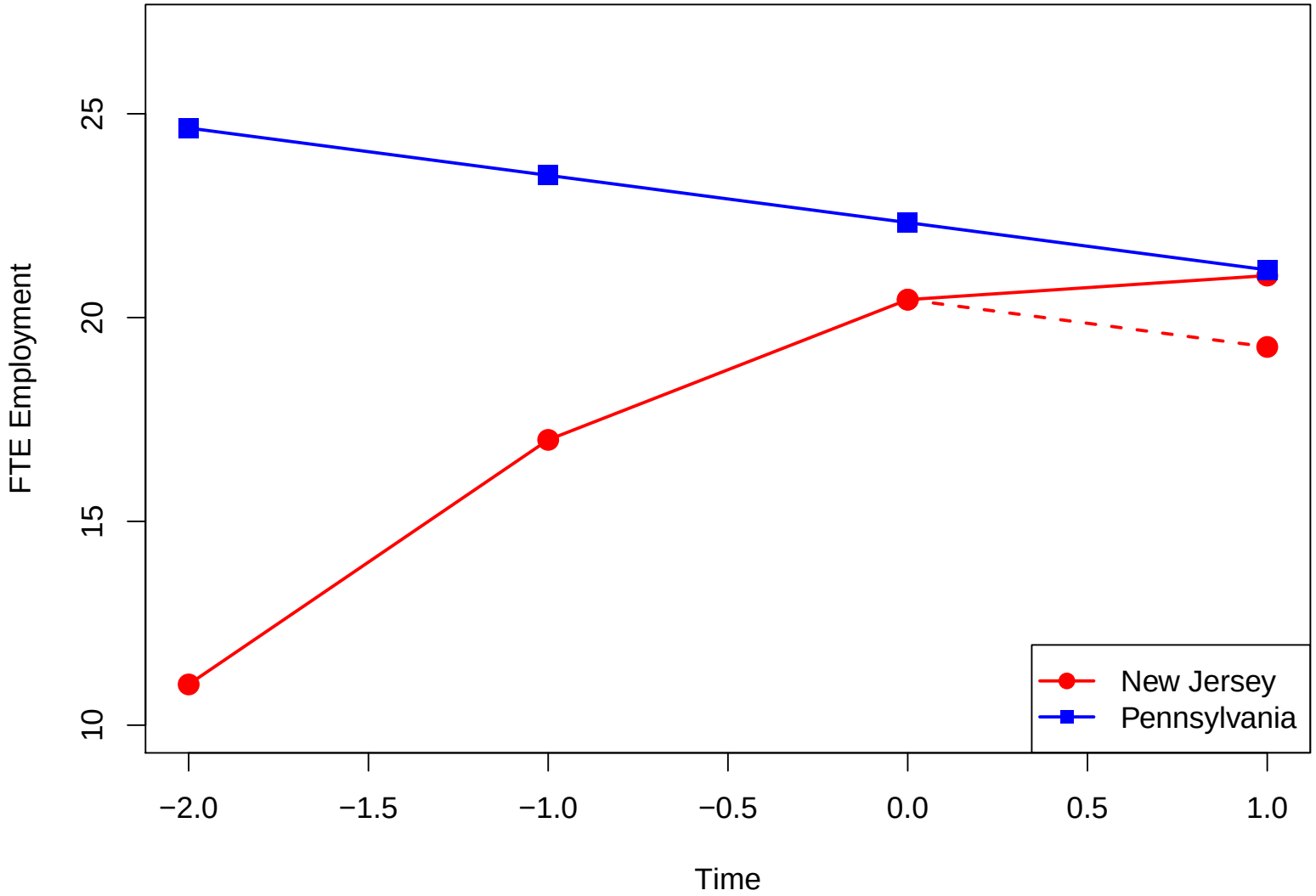
Falsification test: Data for prior periods



Falsification test: Data for prior periods



Falsification test: Data for prior periods



Warning

These are good falsification checks.

But you cannot “prove” you have parallel trends.

So don’t say you can!

Other approaches to worrying about parallel trends

1. **Sensitivity/ partial ID**

   - Recently-best-known is perhaps Rambachan & Roth (2023)
   - Suggests checking if the effect estimate can survive non-parallelness k times larger than the non-parallelness in the prior trends.
   - A lot of papers don't even survive at $k = 1$, which is clearly bad (Chiu, Lan, Liu, Xu, *forthcoming APSR*)
   - Of course there is no (convincing) rule for "how big of a k is big enough"

Other views I wish we had time for:

- Stability-controlled quasi-experiment: reason about shift in $\mathbb{E}[Y(0)|time = 1] - \mathbb{E}[Y(0)|time = 0]$
- Consider pre-tx outcome as a placebo outcome, challenge whether it is really "equiconfounding" as DID suggests.

Threat 2: Functional form dependence

The differencing is not-invariant to changes of scale. Magnitude or sign of the DID effect may be sensitive to this.

- Higher risk when average outcomes for controls and treated are very different at baseline.
- Example: Cash transfers for refugees
  - income of those receiving vouchers rises from 5 to 10
  - over same time, income for comparably skilled group in host community (no vouchers) changes from 20 to 25
  - on original scale, $DID = (10 - 5) - (25 - 20) = 0\%$ effect of voucher
  - but in terms of percentage changes or logs,
    $[100] - [25] = 75$ percentage points
    $[log(10) - log(5)] - [log(25) - log(20)] = log(2) - log(1.25) = .47$
- In general, more similarity in original outcomes is better both for this reason and to make parallel trends assumption more believable

Matching and difference-in-differences

- Parallel trends is more plausible among units that are more similar (both on X and pre-tx Y).
- One strategy: combine matching (or weighting) and DID
  - Match on pre-treatment covariates and perhaps pre-treatment outcomes
  - Run difference-in-differences regression in matched data-set
  - Can also use inverse-propensity score weighting (Hirano, Imbens, and Ridder 2003; Imai and Kim 2012)
  - Ensuring balance on prior outcomes starts to bring you towards other strategies that effectively condition on Y_{pre}, e.g. synthetic control.

Fixed Effects

You could get the same results as DID if you regress the outcome on indicators (fixed effects) for each unit and for time period and an indicator for being treated, i.e.

$$Y_{it} = \alpha_0 + \alpha D_{it} + \gamma_i + \delta_t + \varepsilon_{it}$$

where D is now the interaction of group and time.

This is called two-way fixed effects.

- parallel trends is still the ID assumption

“Modern” DID?

- Everything we’ve done here is to give you a foundation in “classic” DID, meaning two groups, two periods.
- Sometimes you have different units moving into treatment at different times and you still want to imagine something like parallel trends.
- People long used “two-way fixed effect” models in these cases and unfortunately called this DID (?!)
- Then they discovered TWFE does not have a straightforward interepretation akin to DID and is troubled by either biases or confusing interpretations due to:
  - Treatment effect heterogeneity
  - Already-treated as controls
  - Event time misalignment
  - Global vs. local DID assumptions

What to do with staggered treatments etc.?

You can always run a series of separate well-constructed DID for each wave of treatment adoption, understand what each says, then average later.

See Chiu et al. paper cited above for a great review, including 6 estimators, and empirical impact, or the `fect` tutorial on 5 methods at `https://yiqingxu.org/packages/fect/05-panel.html`

An explosion of work has tried to offer different solutions. The main players you'll hear most about are...

Today's hits in place of TWFE

Approach	**What it solves and how**
Sun & Abraham (2021)	Avoids contamination by using cohort-specific event-time indicators; handles heterogeneity and staggered timing via proper reference groups.
Callaway & Sant'Anna (2021)	Computes group-time-specific ATT(g, t) with flexible, nonparametric aggregation; avoids negative weighting and uses only not-yet-treated units as controls.
Goodman-Bacon (2021)	Decomposes TWFE into 2x2 DiD comparisons; shows that bias arises from contaminated comparisons and negative weights.
de Chaisemartin & D'Haultfœuille (2020)	Aggregates clean DiD comparisons using convex weights; produces interpretable estimates even under treatment effect heterogeneity.

But remember...

What's the first-order concern with any of these?

One-way fixed effects

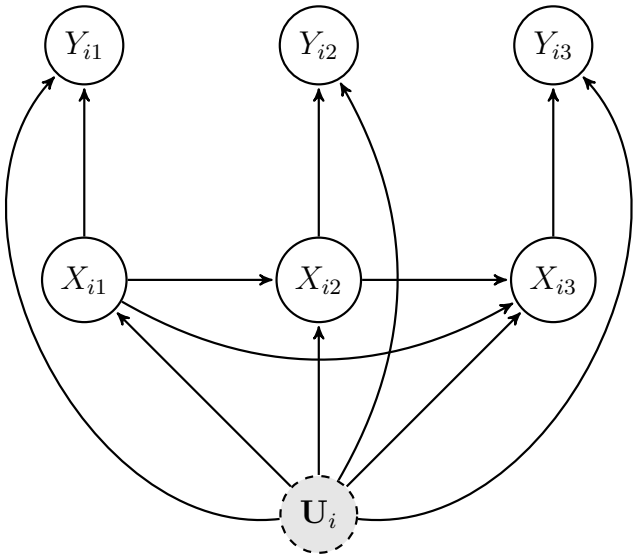
Another set of models use one-way (unit) fixed effects,

$$Y_{it} = \beta_0 + \beta D_{it} + \gamma_i + \varepsilon_{it}$$

- Use when you there are omitted variables, but you think they only a constant shift in the outcome within unit.
- This is not meant to be DID, and applies typically with many periods, units coming in or out of treatment.
- Assumption is stronger than for DID (you aren't allowing any time-related-confounding here)
- e.g. violated if at later time, treatment more likely and non-treatment outcomes tend to be higher.

A closer look at FE

Imai and Kim (2018)



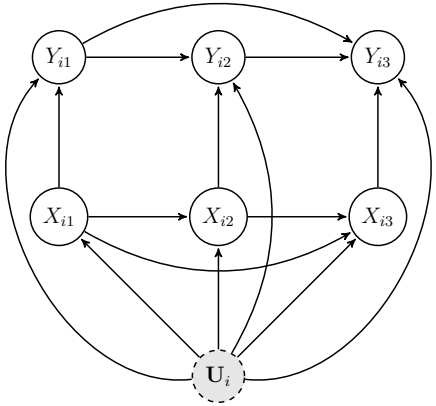
- With unit FEs we will be able to "absorb" U_i, so pretend you can condition on it in these graphs.

Breaks down or requires additional assumptions to manage when:

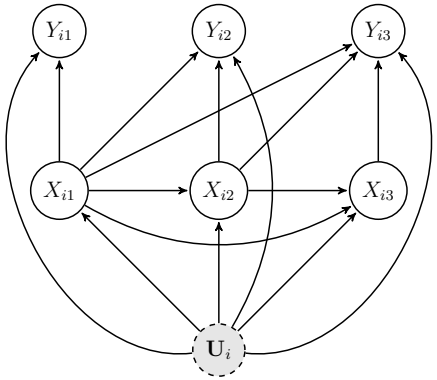
- direct effect of past treatment on current outcome
- direct effect of past outcome on current treatment

Imai and Kim: possible violations

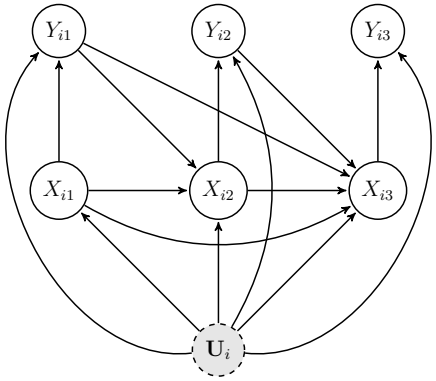
Assume you can condition on past treatments and U_i in the following:



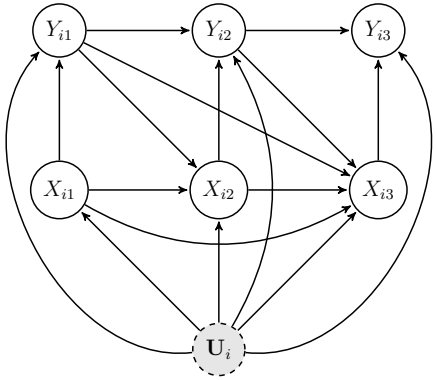
(a) past outcome affects current outcome



(b) past treatments affect current outcome



(c) past outcomes affect current treatment



(d) past outcomes affect both current outcome and treatment

Imai and Kim: possible violations

(a) and (b) are okay, if we can condition on past treatments (and time-invariant characteristics, U_i)

BUT you cannot non-parametrically condition on all prior treatment and on U_i at the same time:

- so instead you must limit the number of lagged treatments that you include (and argue why only these should matter)

(c) violates a core assumption (sequential ignorability) of Imai & Kim by making current treatment depend on prior disturbance.

- Can instead get effect of X_{i3} on Y_{i3} by using X_{i1}, X_{i2}, and Y_{i1} as instruments, while conditioning on U_i and Y_{i2}.
- validity of each of these depends on the assumed absence of its direct causal effect on the outcome Y_{i3}

Wrap-up in the DID vein

One-way FE: use if you can rule out

- all time-varying confounding
- but, the analysis clarifies this is violated by direct effect of past treatment on current outcome, or direct effect of past outcome on current treatment

Diff-in-Diff: Use if you have two periods and...

- you can argue for parallel trends.
- thinking in terms of differences (over time) helps: is the "trend in Y_0 balanced"?
- you can use covariates, if it helps.

Other TSCS approaches, e.g. synthetic control

- allow for more flexibility in terms of number of periods, staggered adoption, heterogeneous treatment effects, non-contemporaneous effects, etc.
- can allow certain types of time-varying confounders
- but still have identification concerns, such as assuming factor structure on confounding and "errors in variables" type problems...a topic for another time.

Bonus: More connections and alternatives

Sequential ignorability:

$$Y_d \perp\!\!\!\perp D | Y_{pre}$$

What is a DAG that would support this?

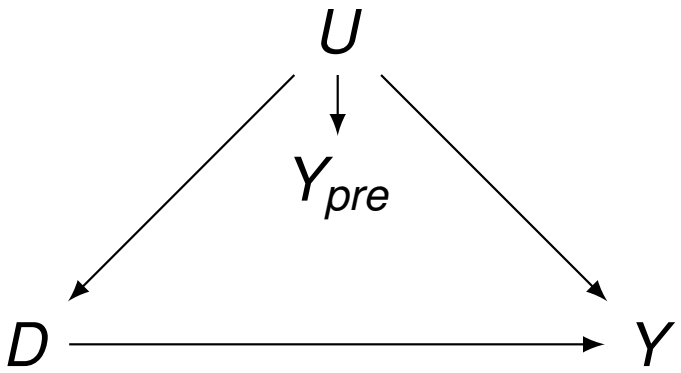
Or if you just cared about the ATT, $Y_0 \perp\!\!\!\perp D | Y_{pre}$

- DID/parallel-trends is not the same, and instead of generally conditioning on Y_{pre}, subtracts it off,

$$Y_0 - Y_{pre} \perp\!\!\!\perp D$$

- If you did really believe SI rather than parallel trends, better to condition on Y_{pre} more flexibly than just subtracting it off, e.g.
  - match on Y_{pre}
  - mean balance on $Y_{pre} \rightarrow$ good if $Y(0)$ affine in $Y_{pre} \rightarrow$ synthetic control/ synthetic DID
  - kernel balance on Y_{pre}: gets $p(Y_{pre}|D = 0)$ close to $p(Y_{pre}|D = 1) \rightarrow$ trajectory balancing

DID as perfect, equiconfounded placebo



The placebo outcome view:

- The pre-treatment outcome is not affected by D so it is perfect
- "parallel trends" is now "equiconfounding" – a parametric claim that on the coefficient scale, Y and Y_{pre} suffer the same strength of confounding
- Can instead argue for equiconfounding on another scale (e.g. R^2)
- Or that confounding with Y is "k times worse" than confounding with Y_{pre}